



Introduction to Reinforcement Learning

Welcome

- CST8509: Reinforcement Learning

- Meet your Professors

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- Contact your Professor
 - email me with enquiries (can expect reply same or next day)
 - email me to arrange office-hour style meetings

Logistics

- Weekly schedule
 - Lecture Mondays 1:00-3:00pm in C346
 - Lab Section 101 Wednesdays 5:00 to 7:00pm in B320
 - Lab Section 102 Wednesdays 7:00 to 9:00pm in B119
 - One hour (average) of asynchronous (Hybrid) activity
- Let's look at the Course Section Information on Brightspace
- Let's look at the submission requirements on Brightspace
- Late lab/assignment submissions are subject to a penalty:
 - 10% < 1 week
 - 100% > 1 week

Expectations for Assignments

Unless an Assignment or Lab explicitly states that it is a group activity, ALL practical and theory work in this course is individual work:

- You must complete solutions by yourself
- You are allowed to participate in study groups and use ChatGPT or similar facilities.
- You are allowed to help each other understand the concepts of the course.
- You are not allowed to copy or use for any purpose any portion of a solution from another student, from ChatGPT, or from any other source.
- You are not allowed to provide any portion of your solution to anyone else.

Tips for Success

Assignments are large bodies of work that cannot reasonably be completed in one or two sessions, even long sessions

Get started early

Get clarification and help early

Make good use of Lab Periods

Tips for Success (cont'd)

Rule One: 15 minute rule

- if you are frustrated and not making progress for 15 minutes, you need to
 - take a break, move on to another part of the assignment, switch to other course work, come back to it later
 - sleep on it
 - seek help from a peer
 - seek help from the instructor

Tips for Success (cont'd)

Rule Two: Don't leave it to the last few days before the due date

- Rule One is not feasible without Rule Two
- Get started early, read through and understand the focus of the assignment and the tasks, as soon as you can
- Keep up with the course pace (every week, you're expected to put in about 5 hours of time in addition to 5 hours of Hybrid Activities, Lectures, and Labs)

Excessive Help

You are trying to increase your OWN knowledge and skill so that you can convince an employer to hire you

Beware of receiving excessive help

Getting the right amount of help when it's needed is important

It's just as important to not get too much help

Do it yourself (you need to learn how to):

- read EVERY word of the Lab and Assignment Documents
- consult course materials and resources
- apply what you read and what you see in videos
- solve apparent inconsistencies/problems

Don't take shortcuts in Labs

Like excessive help, shortcuts are bad

Suppose you are training for a marathon

- your coach wants you to run a five kilometer training route
- You feel overloaded and find a shortcut on the route to reduce it to one kilometer
- You saved so much time doing it your way!
- The short cut took so much less time and effort: where's the problem?

If you cannot explain your own work in a demonstration, you risk getting a zero on the lab or assignment

Plagiarism

Cheating is unacceptable

You may not copy or provide code or text: don't plagiarize

The person you copy from is in just as much trouble as you

Adherence to acceptable standards of academic honesty is an important aspect of the learning process at Algonquin College. Academic work submitted by a student is evaluated on the assumption that the work presented by the student is his or her own, unless designated otherwise. For further details consult Algonquin College Policies AA18(<https://www.algonquincollege.com/policies/files/2016/04/AA18.pdf>) and AA20 (<https://www.algonquincollege.com/policies/files/2016/04/AA20.pdf>)

Overview of this course

- Introduction to RL
- Foundational Principles of RL
 - Mathematical definitions
 - RL Algorithms
- Solving RL problems
 - Game-based, maze based, etc
 - Robotics
 - Create 3
 - OpenAI Gymnasium/Gym, Gazebo

Week 1 Outcomes

What is Reinforcement Learning?

1. Agent
2. Environment
3. Reward
4. Policy
5. Value Function
6. Model

From CST8504: Reinforcement Learning

- Reinforcement Learning is a third type of Machine Learning
- Based on Markov Decision Processes
- Used for agent-based systems
 - Agent uses a policy for choosing an action in each time step
 - Each action taken results in a reward
 - The goal is for the agent to learn a policy that maximizes the reward

Who's who of Reinforcement Learning

- Andrew Barto, University of Massachusetts Amherst
- Richard Sutton, University of Alberta (Co-founder of Edmonton office of DeepMind)
 - textbook
- David Silver, University of Alberta PhD graduate, now at DeepMind and University College London
 - David Silver's RL course on Youtube
 - <https://www.youtube.com/playlist?list=PLzuuYNsE1EZAXYR4FJ75jcJseBmo4KQ9->

How broad and applicable is Reinforcement Learning?

- David Silver lecture 1 (youtube), 6:29
(venn diagrams showing RL is at the intersection of many different fields of human endeavor)

RL vs Supervised and Unsupervised learning

How does reinforcement learning compare to other machine learning (supervised, and unsupervised)?

- David Silver lecture 1 - 9:35

RL Examples

- David Silver Lecture 1: 12:45 - examples of RL, starting with stunt manoeuvres model helicopter (training is done offline with a model of a helicopter world)
- David Silver Lecture 1: after helicopter - video games (many Atari games better than humans) decisions at 15HZ (3 or 4 days of training per game)
- Tetris: <https://www.cbc.ca/player/play/2296942659841>
- our coffee robot?
- monkey and banana?

RL Movie: AlphaGo

- Go board game
- AlphaGo movie
 - begins with Demis Hassabis (co-founder of DeepMind)
 - After Fan Hui, we see David Silver
- At 30:00 the start of the first match

Reinforcement Learning Problem

- Reward is a scalar feedback signal R_t
- R_t represents how well the agent is doing at Step t
- reinforcement learning problems are set up such that goal is to maximize cumulative reward
- Reward Hypothesis: All goals can be described by the maximization of expected cumulative reward

Reward

- reward can be received along the way, or it might come all at the end
- if shorter time is better, then reward per step can be negative, which favors shorter episodes
- to maximize reward overall, agent may need to accept small or negative rewards short-term to maximize the total reward

Anatomy of a Reinforcement Learning Problem

- agent and environment diagram

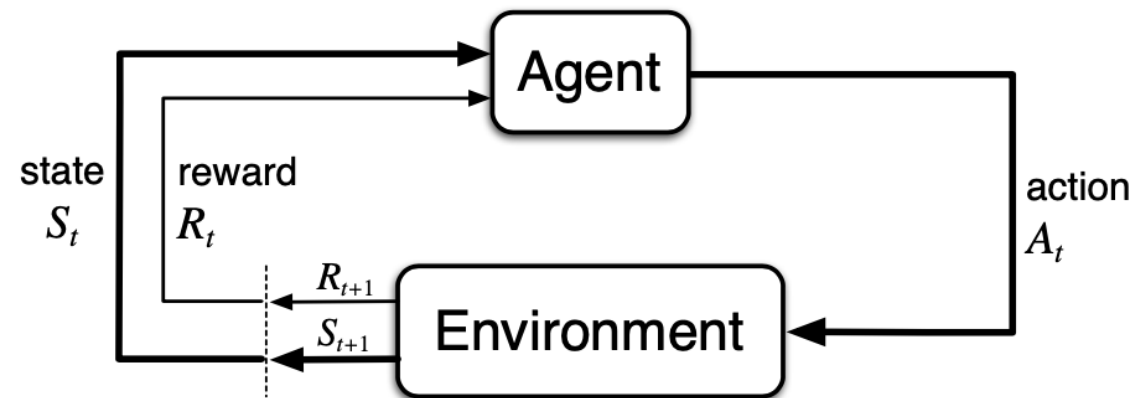


Figure 3.1: The agent–environment interaction in a Markov decision process.

- our algorithm operates in the agent

Anatomy of a Reinforcement Learning Problem

- each time step, the agent receives Reward, Observation, and performs Action
- the time series of Reward, Observation, and Actions is the data for Reinforcement learning
- the history at time step t is $H_t = R_1, O_1, A_1, \dots, R_t, O_t, A_t$

Anatomy of a Reinforcement Learning Problem

- the agent picks the next action at time t based on the information/data contained in H_t
- the environment determines Observations and Rewards
- processing the whole history is cumbersome after many steps
- State is a summary of the information that is used to determine what happens next ($t+1$)

State

- $S_t = f(H_t)$
- environment state (not usually directly accessible by the agent) S_t^e is whatever information is used to pick the next observation and reward
- agent state S_t^a is directly accessible to the agent, the agent keeps track of this, and it's used (somehow) to select the next action
- The programmer is responsible for the (somehow) in the previous point. The programmer decides what the function is:
- $S_t^a = f(H_t)$ for some function f of the programmer's choosing

Markov State

- Important question in RL is whether the state satisfies the Markov Property, in other words, is it a Markov State
- what is the definition of a Markov State?
 - The probability of each possible value for S_t and R_t depends only on the immediately preceding state and action, S_{t-1} and A_{t-1}
- what is the intuition behind a Markov State?
- examples:
 - linear motion of a particle in classical mechanics
 - does the position constitute a Markov state?
 - does the position and velocity constitute a Markov state?

Markov State

- helicopter example (position, velocity, angular velocity, angular position, wind velocity)
- S_t^e environment state is Markov
- H_t is Markov, $S_t = f(H_t) = H_t$
- It's always possible to come up with a Markov state, but we want to identify the Markov states that are more useful for us, efficient, less redundancy, etc.

Rat Example

- David Silver Lecture 1, 47:54
- depending on what function is chosen for $f(H_t)$, the question mark can be electric shock, cheese, or unknown.

RL Agents

- RL Agents may include one or more of the following
 - Policy: function that maps state to action
 - Value Function: represents the value (how good is it?) of each state or action
 - Model: agent's internal representation of the environment, as opposed to the environment itself

Policy

- Function: map from state to action
- deterministic policy is one where there is only one choice, one action
 - $a = \pi(S)$
- our goal will be to learn a function π , from experience, such that we maximize reward
- stochastic policy
 - $\pi(a|s) = P[A=a|S=s]$
 - This function gives the probability of one or more actions, given State s (non-deterministic)
 - Example: in certain state, a_1 chosen 20% of the time, a_2 chosen 80% of the time

Value Function

- the value function indicates how good it is to be in a particular state with respect to expected future reward
- used to pick actions, interacts with policy
- the value function for a policy is the sum of the expected reward for all future states

Value Function (cont'd)

- in the case of Atari games, as states are visited, value goes up and down, because
 - if something good is about to happen, value function is elevated
 - after something good happens, that reward is behind you, and not included in the future reward
 - in other words the value function does not include the sum of rewards received so far, just future reward, which oscillates: David Silver Lecture 1: 1:02:15

Model

- allows inferences about how the environment will behave
- model indicates or implies the next state and next reward.
- From Sutton, page 7: "Models are used for planning, by which we mean any way of deciding on a course of action by considering possible future situations before they are actually experienced."
- David Silver Lecture 1: Transition model (predicts states) Reward Model (predicts rewards)
- not all Reinforcement Learning Problems/Solutions include a model

Maze Example

- David Silver, Lecture 1: 1:08:00

Taxonomy of RL Agents

David Silver Lecture 1:11:0

- Value Based
 - no policy (choose actions based on Value function)
 - value function
- Policy Based
 - Policy
 - no value function

Taxonomy of RL Agents

- Actor Critic
 - Policy (actor)
 - Value Function (critic)
- Model Free
 - Policy and/or Value Function
 - no Model
- Model Based
 - Policy and/or Value Function
 - Model

Key Subproblems

Learning vs Planning, David Silver Lecture 1: 1:16:10

- reinforcement learning
- planning
- balancing exploitation and exploration
 - Example:
 - Always go to a good restaurant (exploit the good restaurant)
 - Randomly choose a new restaurant (exploration, might be better, might be worse. This is our chance to find better, but it's a risk)
- prediction (evaluate future reward) vs control (optimize policy)

Time to check your learning!

Let's see how many key concepts you recall by answering the following questions!

- What is the Markov Property?
- What are the possible components in an RL agent?
- What is a policy in the context of RL?
- What is a value function in the context of RL?
- What is a model in the context of RL?